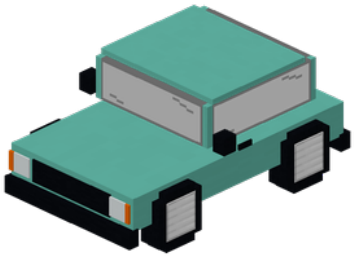
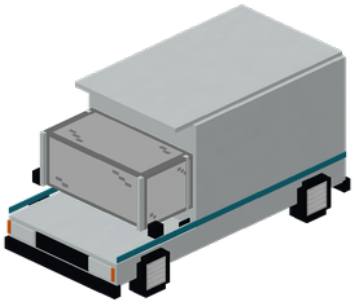
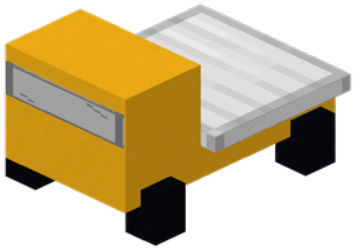
NEVERGATE / MINECRAFT JAVA

NG-Fahrzeuge

User manual & complete illustrated catalogue

0.1.0-dev12 · 17.10.2026

39 entries with original model images, usage and recipes.

5 categories and 16 practical guides.

Minecraft Java 1.20.2 · Forge 48.1.0 · Java 17.

The completed development build is active on MC02. Later work still in progress is not documented as completed.

Images show static inventory models. Dynamic screen text, motion and lighting effects may appear differently in game.

Guides

01 Installation and current version
02 Driving, displays and comfort
03 Build and buy vehicles
04 Keys, ownership and cargo
05 Fuel and refuelling
06 Damage, repair and lift
07 Coupling and cargo trailer
08 Garage and registration plates
09 Towing and recovery
10 Battery and jump starting
11 Checks and limits
12 Equipment, tyres and vehicle overview
13 Bus, motorcycle and pickup
14 All-wheel drive, transmission, taxi and doors
15 Tractor unit and 108-slot semitrailer
16 Cooling and workshop diagnostics

The full catalogue follows, grouped by category. For shaped recipes, / separates crafting rows and · means an empty slot.

01 / Installation and current version

Minecraft Java 1.20.2 · Forge 48.1.0 · Java 17. MC02 runs 0.1.0-dev12.

1. Close Minecraft, back up the world and remove older active copies of the same mod.

2. Use the same current file on client and server: ng_fahrzeuge-0.1.0-dev12.jar.

3. NG Traffic beta2-dev12 and NG Vehicles dev12 are separate mods. Install both matching JARs to use them together.

4. The catalogue contains 39 items, including eleven vehicle types and additional variants.

5. This build is active on MC02. Development/test build, not a stable full release.

02 / Driving, displays and comfort

Only the driver controls the vehicle.

1. I engine, H parking brake, L lights, comma/period indicators, R hazards, G horn, B cargo.

2. O opens doors, U opens the rear: engine off and parking brake on. Starting the engine closes doors.

3. C sets cruise control while driving forward. Braking, reversing, engine off or collision cancels it.

4. K wipers, V power windows, N cabin light. Open doors switch the cabin light on automatically.

5. The HUD shows speed, fuel, condition, battery and total/trip distance. One block equals one metre. Mirrors are model surfaces; lamps do not cast dynamic light beams.

03 / Build and buy vehicles

The vehicle workbench and dealer are interactive blocks.

1. Craft and right-click the workbench, then select a vehicle. Check costs and a free inventory slot; materials are deducted server-side.

2. Right-click the dealer, inspect preview/price and buy. Defaults: City 24, Family 40, Transporter 56, tow truck 96 emeralds.

3. Dealer prices and payment item are configurable under dealer in ng_fahrzeuge-server.toml. Payment uses items; euros are not included.

4. Roof-box Family and windowed Transporter have separate items and recipes. Their variant is retained when picked up or stored.

04 / Keys, ownership and cargo

The placing player becomes owner; authorised keys allow additional drivers.

1. Right-click your vehicle with a blank key to bind it. Right-click with a bound key to lock/unlock an unoccupied vehicle.

2. Keep the key in your inventory when entering. Cargo requires engine off and parking brake on; use B or sneak-right-click with a matching key.

3. Cargo capacity: City 9, Family 27, Transporter 54 slots. Contents are saved.

4. Vehicle papers open management: name, plate, doors, trip distance, coupling, key revocation and ownership transfer. Changing owner invalidates old keys.

5. To pick up: unoccupied, with no open cargo screens or connections; sneak-right-click with an empty hand. Vehicle data stays on the item.

05 / Fuel and refuelling

Fuel is a fictional game item.

1. Right-click an authorised parked vehicle with a filled can. Engine must be off and parking brake on. A can holds 10 l.

2. A can + coal + blaze powder refills to 10 l; preferably use an empty can.

3. Fill the pump with a can, up to 500 l. Right-click without a can to refuel the nearest authorised vehicle within six blocks.

4. City 40 l / 6.5 l per 100 km; Family 55 l / 8 l; Transporter 80 l / 12 l. Idling also consumes fuel.

06 / Damage, repair and lift

Body and engine condition are stored separately.

1. Hard collisions cause damage. Severe engine damage limits power; a broken vehicle will not start.

2. Right-click with the repair tool while carrying spare parts. Each repair uses one part and one durability point, restoring up to 25 points to each condition value.

3. Place the lift next to an authorised parked vehicle and right-click. Use the repair tool to service it; another item lowers it.

4. Raising/lowering requires free space. If the lowering area is blocked, the vehicle remains raised.

07 / Coupling and cargo trailer

Family and Transporter can tow the cargo trailer; City cannot.

1. Place the trailer behind the tow vehicle. Engine off and parking brake on; use J or management to couple.

2. The trailer must be owned or authorised and have no inventory screen open.

3. Right-click to open its 27 cargo slots. A blocked trailer stops the towing vehicle engine.

4. Pick up only uncoupled trailers: sneak-right-click with an empty hand. Connections persist without loading distant chunks.

08 / Garage and registration plates

Vehicle papers open management for an authorised parked vehicle.

1. Set a plate with up to twelve letters, digits, spaces and hyphens; it appears front and rear.

2. Right-click the garage. Store and retrieve up to nine owned vehicles per player across the server.

3. Occupants, open cargo screens, trailers or recovery connections prevent storage. UUID, keys, paint, cargo and condition are retained.

4. The area in front of the garage must be clear and loaded. Failed retrieval retains the stored record.

09 / Towing and recovery

Access to both vehicles is required.

1. Tow rope: right-click the tow vehicle, then the broken-down vehicle. Both must be stopped, engine off and parking brake on. Sneak-right-click disconnects.

2. Sneak-right-click the tow truck with the recovery remote to toggle the ramp. Normal right-click loads the nearest authorised vehicle; click again to unload behind it.

3. Occupied broken-down vehicles cannot be loaded. A blocked unloading area prevents unloading.

4. The recovered vehicle engine is disabled during recovery. Ownership/key changes are blocked; distant chunks are not loaded.

10 / Battery and jump starting

Battery charge is shown in the HUD.

1. Starting needs at least 5% and consumes two percentage points. A running engine charges the battery.

2. Lights, cabin lighting and wipers consume charge while parked; power windows require charge.

3. Jumper cables: right-click donor, then receiver. Maximum distance eight blocks; both vehicles must be accessible.

4. Donor at least 20%, receiver below 20%: ten percentage points are transferred.

11 / Checks and limits

As of 17 October 2026: Vehicles dev12 and Traffic beta2-dev12 are loaded on MC02.

1. Vehicles: 129 server tests passed again. Traffic/Vehicles: 16 targeted integration tests, including state access for all eleven vehicle types.

2. Separate server processes confirm state/receipt persistence and replay protection; actual chunk unloading and return were checked.

3. Historical client and two-client evidence applies to its documented builds. All dev12 features with two real drivers and every semitrailer parking geometry still need testing.

4. Electric drive and economy provider are missing. Paid external fuel/charging/wash services remain blocked.

5. Author: NeverGate. Existing rights notices remain in force; original data and recipes come from the current JARs.

12 / Equipment, tyres and vehicle overview

Dev10 and dev11 equipment is included in dev12.

1. F6 changes vehicle camera. M opens the bonnet and Z the roof inventory; roof racks have 18 slots.

2. For tyre replacement, stop the engine, apply parking brake and use jack and spare wheel.

3. Operate paint and wash stations only with access to the parked vehicle. Dirt and paint are saved.

4. My vehicles opens type/location overview and tracking. It does not load distant chunks to search for vehicles.

13 / Bus, motorcycle and pickup

Additional vehicle types use the same ownership and driving foundation.

1. City bus: driver and twelve passengers, separate doors, F9 stop request and configurable route displays.

2. Motorcycle: rider and pillion. Operate the side stand with F8 and raise it before riding. There are no doors.

3. Pickup: tailgate and bed with 27 cargo slots. Load and unload while safely parked.

4. F7 controls the belt on suitable vehicles. L cycles light levels; F10 high beam and F12 fog lights. All key bindings can be changed.

14 / All-wheel drive, transmission, taxi and doors

The new controls are also available in the Mechanics tab of vehicle management.

1. Off-roader: numpad 4 all-wheel drive, 5 low range and 6 hill hold. Change drive settings while stopped with parking brake on; low range requires all-wheel drive.

2. Numpad 7 changes automatic/manual mode; + and - shift gears. Change forward/reverse through neutral only when nearly stationary.

3. Watch gear and RPM in the HUD. A gear that is too low limits speed.

4. Numpad 8 controls the taxi sign and 9 the cargo lock. Front-left/right and rear-left/right doors use 0/1/2/3. Close doors before driving.

5. Taxi: free when empty or driver only, occupied with passengers. Remote keys operate within 32 blocks on loaded authorised targets.

15 / Tractor unit and 108-slot semitrailer

Tractor unit and semitrailer must be aligned correctly to couple.

1. Place the trailer behind the tractor with supports down and align kingpin over fifth wheel. Press J to couple while safely parked.

2. Raise supports using the crank. Only then start and drive. Obstacles and articulation limit movement; allow room when manoeuvring.

3. Right-click opens cargo. The page button switches between slots 1-54 and 55-108 of the same saved inventory.

4. To park: engine off, parking brake on, supports down, then J to uncouple. Open cargo access blocks support changes/uncoupling.

5. Pick up an uncoupled trailer by sneaking with an empty hand; cargo and support state persist. Not every semitrailer parking geometry has been fully tested.

16 / Cooling and workshop diagnostics

The cooling circuit holds 5 L; a new coolant can holds 4 L.

1. Stop the engine, apply parking brake, open bonnet and wait until temperature is below 60 °C.

2. Use the coolant can to add at most 1 L per click; only the missing amount is consumed.

3. Damaged radiators leak coolant while the engine runs. Power drops above 105 °C, engine damage starts above 115 °C and the engine stops at 130 °C.

4. Open the diagnostic scanner on the parked vehicle. Inspect body, engine, radiator and tyres; carry the required tools and spare parts.

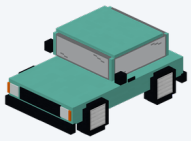
5. Existing repair functions also restore the radiator. Severe impacts automatically activate hazard lights.

Complete catalogue

39 registered items and blocks. Recipes come from mod resources; vehicle assembly costs come from the vehicle workbench.

Vehicles and variants / 13

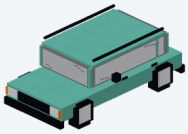
NG City



Vehicle workbench
24 × Iron ingot · 8 × Copper ingot · 6 × Glass · 4 × Black wool · 4 × Redstone
Output: 1

Single-seat city car, 9 cargo slots and 40-l tank. Right-click ground to place; I starts, H releases parking brake. Not approved for trailers.

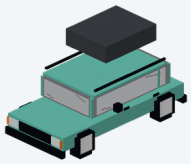
NG Family



Vehicle workbench
36 × Iron ingot · 12 × Copper ingot · 8 × Glass · 4 × Black wool · 6 × Redstone
Output: 1

Four-seat family car with 27 cargo slots and 55-l tank. Supports cargo trailers; I engine, H parking brake.

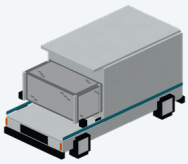
NG Family · Roof box



Crafting table
3 × Iron block · 2 × Glass · 1 × Chest · 2 × Black wool · 1 × Redstone block
Output: 1
Grid: III / GEG / WRW
I=Iron block; G=Glass; E=Chest; W=Black wool; R=Redstone block

Family variant with a visible roof box. Separate item/recipe; variant persists through pickup and storage.

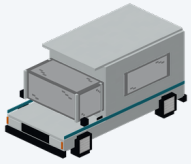
NG Transporter



Vehicle workbench
48 × Iron ingot · 16 × Copper ingot · 10 × Glass · 4 × Black wool · 8 × Redstone
Output: 1

Two-seat van with 54 cargo slots, 80-l tank and trailer coupling. White base paint with mint stripe.

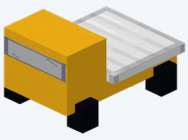
NG Transporter · Windows



Crafting table
3 × Iron block · 2 × Glass · 1 × Glass · 2 × Black wool · 1 × Redstone block
Output: 1
Grid: III / GEG / WRW
I=Iron block; G=Glass; E=Glass; W=Black wool; R=Redstone block

Transporter variant with side cargo-area windows. Separate item/recipe; variant is saved.

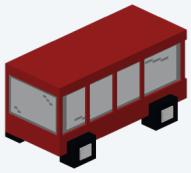
NG Tow Truck



Vehicle workbench
60 × Iron ingot · 20 × Copper ingot · 12 × Glass · 4 × Black wool · 10 × Redstone
Output: 1

Two-seat tow truck with flatbed and ramp. Load/unload vehicles with the recovery remote; check permissions and clear space.

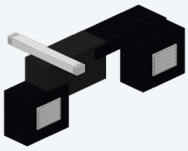
NG City Bus



Vehicle workbench
72 × Iron ingot · 24 × Copper ingot · 14 × Glass · 4 × Black wool · 12 × Redstone
Output: 1

City bus with driver and twelve passengers, separate bus doors, stop request and route displays. Tank: 180 L. Configure route and destination in vehicle management.

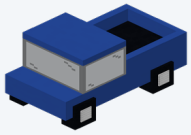
NG Motorcycle



Vehicle workbench
84 × Iron ingot · 28 × Copper ingot · 16 × Glass · 4 × Black wool · 14 × Redstone
Output: 1

Motorcycle with rider and pillion seat, side stand and lean. Tank: 18 L. Raise the stand before riding; there are no doors.

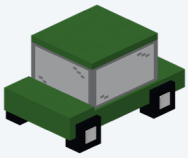
NG Pickup



Vehicle workbench
96 × Iron ingot · 32 × Copper ingot · 18 × Glass · 4 × Black wool · 16 × Redstone
Output: 1

Two-seat pickup with open bed, folding tailgate and 27 cargo slots. Tank: 80 L. Manage cargo and rear access while safely parked.

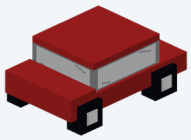
NG Off-roader



Vehicle workbench
108 × Iron ingot · 36 × Copper ingot · 20 × Glass · 4 × Black wool · 18 × Redstone
Output: 1

Four-seat off-roader with 85 L tank and increased clearance. Change all-wheel drive and low range while stopped with parking brake on; hill hold assists short stops.

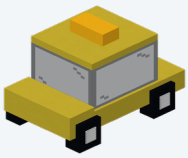
NG Sports Car



Vehicle workbench
120 × Iron ingot · 40 × Copper ingot · 22 × Glass · 4 × Black wool · 20 × Redstone
Output: 1

Low sports car with two seats and a 60 L tank. Use automatic or six-speed manual transmission and watch the gear/RPM display.

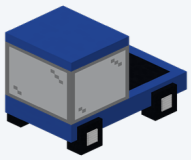
NG Taxi



Vehicle workbench
132 × Iron ingot · 44 × Copper ingot · 24 × Glass · 4 × Black wool · 22 × Redstone
Output: 1

Taxi with driver and three passengers and a 55 L tank. The roof sign indicates free/occupied according to passengers and can be disabled separately.

NG Tractor Unit

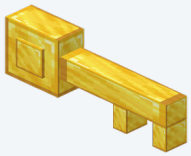


Vehicle workbench
144 × Iron ingot · 48 × Copper ingot · 26 × Glass · 4 × Black wool · 24 × Redstone
Output: 1

Two-seat tractor unit with six wheels and a 240 L tank. Couple the box semitrailer, raise its supports and close doors before starting and driving.

Keys and management / 3

Vehicle key



Crafting table
1 × Gold nugget · 2 × Iron nugget
Output: 1
Grid: ·G / II
G=Gold nugget; I=Iron nugget

Bind a blank key to your vehicle. A valid bound key also locks/unlocks within 32 blocks; sneak-use toggles the separate cargo lock. The target must be loaded.

Vehicle papers



Crafting table
3 × Paper · 1 × Redstone
Output: 1
Grid: PPP / ·R·
P=Paper; R=Redstone

Open management on an authorised parked vehicle: name, plate, doors, distance, coupling, keys and owner. The Mechanics tab provides the new dev12 controls.

My vehicles



Crafting table
1 × Paper · 2 × Iron ingot · 1 × Compass · 1 × Redstone
Output: 1
Grid: ·P· / ICI / ·R·
P=Paper; I=Iron ingot; C=Compass; R=Redstone

Open your vehicle overview with type and last known location. Tracking points to the stored target; distant chunks are not automatically loaded.

Assembly, dealers and workshop / 6

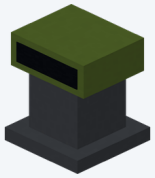
Vehicle workbench



Crafting table
3 × Iron ingot · 2 × Redstone · 1 × Crafting table · 3 × Wood planks (any)
Output: 1
Grid: III / RCR / PPP
I=Iron ingot; R=Redstone; C=Crafting table; P=Wood planks (any)

All eleven vehicle types are available for assembly. The listed materials are checked in full and a free inventory slot is required. Some variants use separate crafting-table recipes.

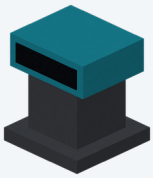
NG Vehicle dealer



Crafting table
3 × Glass · 5 × Iron ingot · 1 × Emerald
Output: 1
Grid: GGG / IRI / III
G=Glass; I=Iron ingot; R=Emerald

Select a vehicle, inspect preview and price and buy using the configured inventory item, emeralds by default. Selling uses the displayed return value; no euro payments.

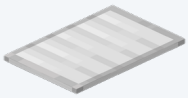
NG Garage



Crafting table
8 × Iron ingot · 1 × Chest
Output: 1
Grid: III / ICI / III
I=Iron ingot; C=Chest

Store/retrieve up to nine owned vehicles per player server-wide. Keep the exit clear; identity, keys, cargo and condition persist.

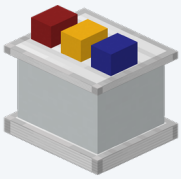
NG Workshop lift



Crafting table
6 × Iron ingot · 1 × Piston · 1 × Redstone
Output: 1
Grid: I·I / IPI / IRI
I=Iron ingot; P=Piston; R=Redstone

Place next to an authorised parked vehicle and right-click. Service with the repair tool; another item lowers it when space is clear.

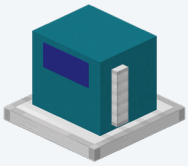
Paint station



Crafting table
2 × Red dye · 1 × Green dye · 1 × Blue dye · 5 × Iron ingot
Output: 1
Grid: RGB / III / IRI
R=Red dye; G=Green dye; B=Blue dye; I=Iron ingot

Place next to the parked vehicle. Check dye and access permissions and select the desired paint through the station.

NG Wash Station

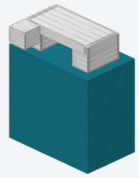


Crafting table
7 × Iron ingot · 1 × Water bucket · 1 × Redstone
Output: 1
Grid: III / IBI / IRI
I=Iron ingot; B=Water bucket; R=Redstone

Wash station for the existing dirt system. Keep the vehicle authorised and parked and operate the wash at the station.

Fuel, cooling and repairs / 8

Fuel can (10 l)



Crafting table
3 × Iron ingot · 1 × Coal · 1 × Blaze powder
Output: 1
Grid: ·I· / ICI / ·B·
I=Iron ingot; C=Coal; B=Blaze powder

Holds 10 l of fuel. Right-click an authorised parked vehicle or pump; can + coal + blaze powder refills it.

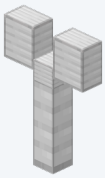
NG Fuel pump



Crafting table
5 × Iron ingot · 1 × Glass · 2 × Copper ingot · 1 × Redstone
Output: 1
Grid: IGI / CRC / III
I=Iron ingot; G=Glass; C=Copper ingot; R=Redstone

Stores up to 500 l. Fill with a can; right-click without a can to refuel the nearest authorised vehicle within six blocks.

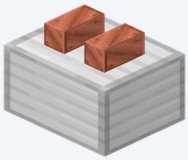
Repair wrench



Crafting table
3 × Iron ingot · 2 × Stick
Output: 1
Grid: ·II / ·SI / S··
I=Iron ingot; S=Stick

Right-click an authorised parked vehicle while carrying spare parts. Restores up to 25 points to each condition value.

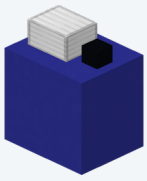
Vehicle spare parts



Crafting table
2 × Iron ingot · 1 × Copper ingot · 1 × Redstone
Output: 4
Grid: ICI / ·R·
I=Iron ingot; C=Copper ingot; R=Redstone

Carry alongside the repair tool for repairs. Each repair consumes one spare part.

Coolant Can (4 L)



Crafting table
5 × Iron ingot · 1 × Blue dye · 1 × Water bucket
Output: 1
Grid: IBI / IWI / ·I·
I=Iron ingot; B=Blue dye; W=Water bucket

Holds 4 L coolant. Use with engine off, parking brake on, bonnet open and temperature below 60 °C. Add at most 1 L per click up to the 5 L cooling capacity.

Workshop Diagnostic Scanner



Crafting table
5 × Iron ingot · 1 × Redstone · 2 × Glass · 1 × Redstone comparator
Output: 1
Grid: IRI / GCG / III
I=Iron ingot; R=Redstone; G=Glass; C=Redstone comparator

Right-click an authorised parked vehicle. Inspect body, engine, radiator and tyres and the required spare parts/tools in the workshop view.

Vehicle Jack



Crafting table
4 × Iron ingot · 1 × Piston
Output: 1
Grid: ·I· / IPI / ·I·
I=Iron ingot; P=Piston

Use for wheel replacement on an authorised parked vehicle. Stop the engine, apply the parking brake and carry a spare wheel.

Spare Wheel

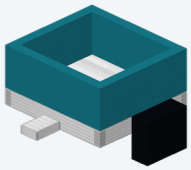


Crafting table
8 × Black wool · 1 × Iron ingot
Output: 1
Grid: WWW / WIW / WWW
W=Black wool; I=Iron ingot

Carry as a replacement for damaged tyres. Use the jack to change a wheel on a safely parked vehicle.

Trailers and recovery / 9

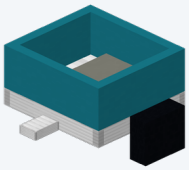
NG Cargo trailer



Crafting table
4 × Iron ingot · 1 × Chest · 3 × Wood planks (any)
Output: 1
Grid: ICI / PPP / I·I
I=Iron ingot; C=Chest; P=Wood planks (any)

Trailer with 27 cargo slots. Place behind Family/Transporter and couple using J or management; pick up only while uncoupled.

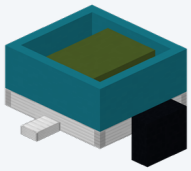
NG Box Trailer



Crafting table
5 × Iron ingot · 1 × Chest · 2 × Black wool
Output: 1
Grid: III / ICI / W·W
I=Iron ingot; C=Chest; W=Black wool

Enclosed box trailer based on the cargo trailer. Place behind a suitable towing vehicle and couple using J or vehicle management.

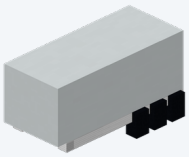
NG Canvas Trailer



Crafting table
3 × Green wool · 2 × Iron block · 1 × Chest · 2 × Black wool
Output: 1
Grid: WWW / ICI / B·B
W=Green wool; I=Iron block; C=Chest; B=Black wool

Canvas variant of the cargo trailer. Cargo is saved; couple only while safely parked and pick up only when uncoupled.

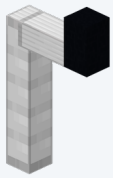
Box Semitrailer



Crafting table
3 × Iron block · 2 × Chest · 1 × Glass · 3 × Black wool
Output: 1
Grid: III / CGC / WWW
I=Iron block; C=Chest; G=Glass; W=Black wool

Box semitrailer with 108 cargo slots on two pages of 54. Align kingpin with fifth wheel; check supports and leave room for manoeuvres.

Trailer Support Crank



Crafting table
4 × Iron ingot
Output: 1
Grid: ·II / ·I· / ·I·
I=Iron ingot

Right-click the parked semitrailer to lower/raise supports. Supports must be raised for driving and lowered before uncoupling.

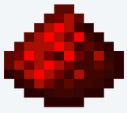
Tow rope



Crafting table
5 × String
Output: 1
Grid: S·S / ·S· / S·S
S=String

Click towing vehicle, then disabled vehicle. Both must be stopped, engine off and parking brake on. Sneak-right-click disconnects.

Jumper cables



Crafting table
2 × Iron ingot · 1 × Redstone · 1 × Copper ingot
Output: 1
Grid: IRI / ·C·
I=Iron ingot; R=Redstone; C=Copper ingot

Click donor and receiver within eight blocks. If donor is at least 20% and receiver below 20%, ten points are transferred.

Tow truck remote



Crafting table
3 × Iron ingot · 1 × Redstone · 1 × Copper ingot
Output: 1
Grid: ·R· / ICI / ·I·
I=Iron ingot; R=Redstone; C=Copper ingot

Right-click the tow truck to load/unload; sneak-right-click toggles the ramp. Access to both vehicles is required.

Roof Rack



Crafting table
5 × Iron bars
Output: 1
Grid: III / I·I
I=Iron bars

Equip an authorised parked vehicle with a roof rack. The saved roof inventory has 18 slots and opens with Z.

Sources and notices

https://github.com/InDance2018/NG-Fahrzeuge

https://nevergate.de/ng-fahrzeuge/

Minecraft 1.20.2 / Forge 48.1.0

Model and recipe data: 0.1.0-dev12

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